Mathematics (3)
(MTH203,MTHN103,MTHn03)
Revision sheet

Differential Equations and Transforms

Assoc.Prof/ Ashraf Taha El-Sayed Taha

Dr/ Amira Mohammed Salem

MTH203

[1] Find the Laplace transform of the following functions:

(a) $f(t)=3e^{-2t}+5t^{8}$.

(b) $f(t)=3\sin^{2}t+\sinh 5t$.

(c) $f(t)=4\sin 4t\sin 3t+3\cosh\sqrt{7}t$.

(d) $f(t)=\cos^{2}6t+\cos 3t\cos 2t$.

(e) $f(t)=\sin 4t\cos 3t+\sin 3t\cos 3t$.

(f) $f(t)=3e^{5t}\left[\sin^{2}t+4\sin t\cos 2t\right]$

(g) $f(t)=3e^{-2t}t^{7}+\cosh\sqrt{3}t+t\sin t$.

(h) $f(t)=u_{3}(t)e^{4(t-3)}(4t-1)$

(i) $f(t)=u_{5}(t)\cos 2(t-5)+\dfrac{\sin t}{t}$.

(j) $f(t)=\int_{0}^{t}7\tau^{3}+2\cos 4\tau\ \ d\tau$.

[2] Find the inverse Laplace transform of the following functions:

(a) $F(s)=\dfrac{4s+3}{s^{2}+64}+\dfrac{3}{s^{7}}$.

(b) $F(s)=\dfrac{s-4}{s(s-3)(s+2)}$.

(c) $F(s)=\dfrac{7}{s^2+4s+6}$.

(d) $F(s)=\dfrac{s}{s^2+10s-3}$.

(e) $F(s)=\dfrac{4se^{-2s}}{s^2+16}+\dfrac{e^{-7s}}{s+3}$.

(f) $F(s)=\dfrac{(s+9)e^{-6s}}{s^2+6s+3}$

(g) $F(s)=\ln(s^2+25)$.

(h) $F(s)=\tan^{-1}\left(\dfrac{s}{4}\right)$.

[3] Find the solution of the following differential equations using the Laplace transform:

(a) $y''-16y=0, \quad y(0)=1, y'(0)=4$.

(b) $y''+5y'+3y=0, \quad y(0)=3, y'(0)=1$.

(c) $y'-4y=8, \quad y(0)=2$.

[4] Expand the periodic function

(a) $f(x)=x^3, \ -\pi\le x\le\pi$ into a Fourier series.

(b) $f(x)=|x|, \ -\pi\le x\le\pi$ into a Fourier series.

(c) $f(x)=4x, \ 0\le x\le 3$ into a Fourier cosine series.

(d) $f(x)=5x^2, \ 0\le x\le 2$ into a Fourier sine series .

MTH203

Good luck